Happiest Health

Digital Twin

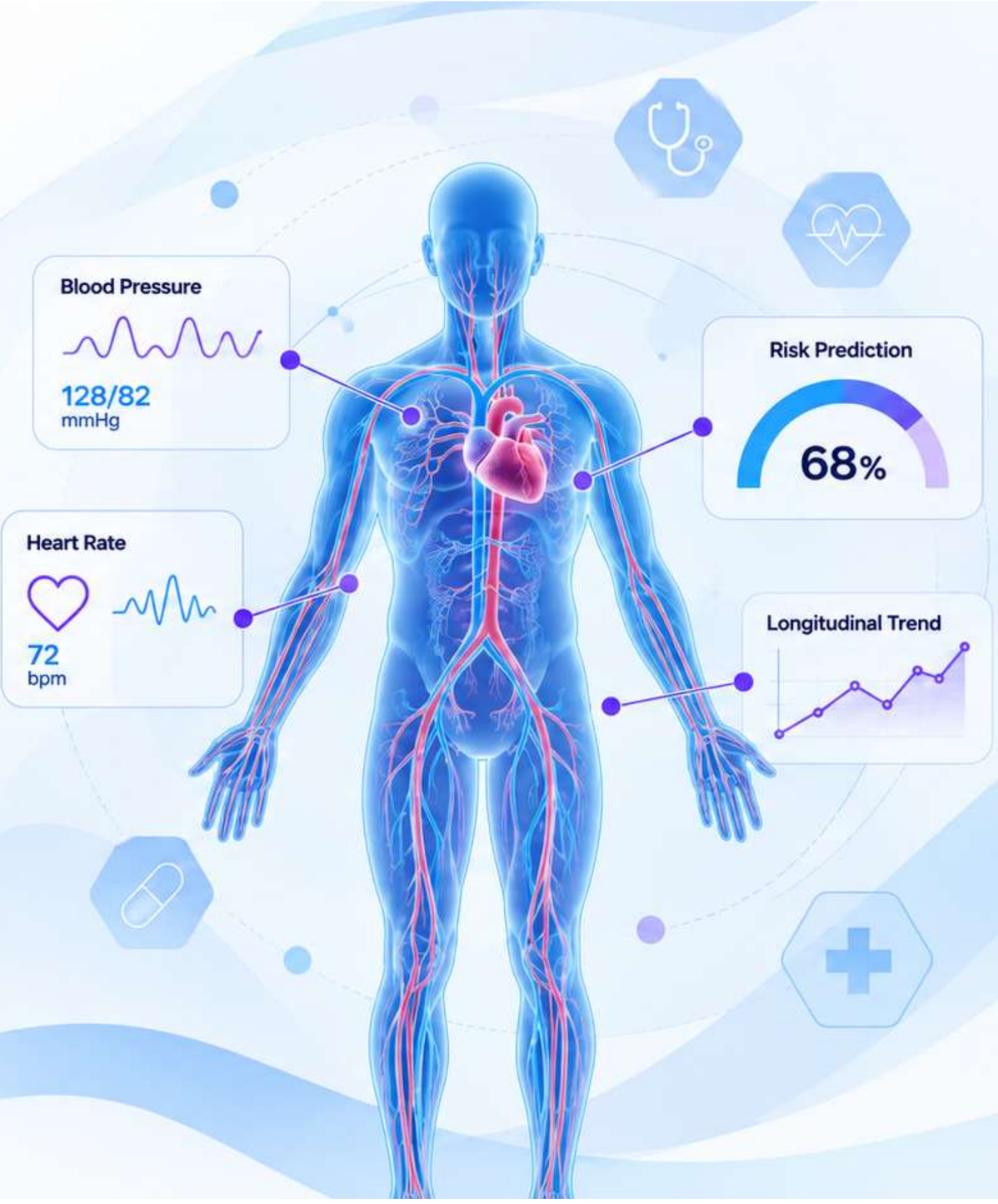
CHALLENGE

Build. Predict. Reimagine Healthcare.

VitalTwin

AI-Powered Clinical Digital Twin

A Living Digital Perspective on Health



Longitudinal Patient Data

AI Risk Prediction

Explainable Insights

What-If Simulation

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National Institute of Technology Rourkela

Happiest Health Digital Twin Challenge 2026

Reimagining and Reforming Healthcare in India Summit 2026, Bengaluru

VitalTwin — AI-Powered Clinical Digital Twin

Shifting Cardiovascular Care from Reactive to Proactive



Objective

Shift from reactive crisis response to 365-day continuous prevention.

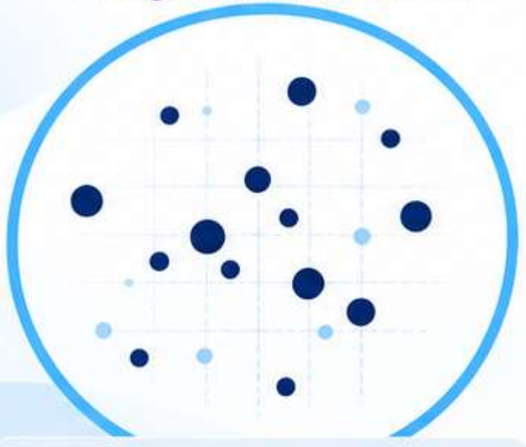
Target Problem

Hypertension is a major and often under-monitored chronic condition.

Users & Value

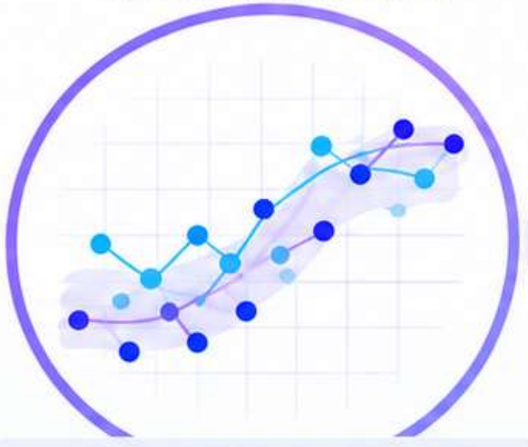
For primary care clinicians and patients. Delivers actionable, explainable insights to support earlier risk identification.

Scattered Longitudinal Records



Isolated, traditional records with no continuous tracking.

Digital Twin State Generation



Creation of a longitudinal patient-state representation from historical data.

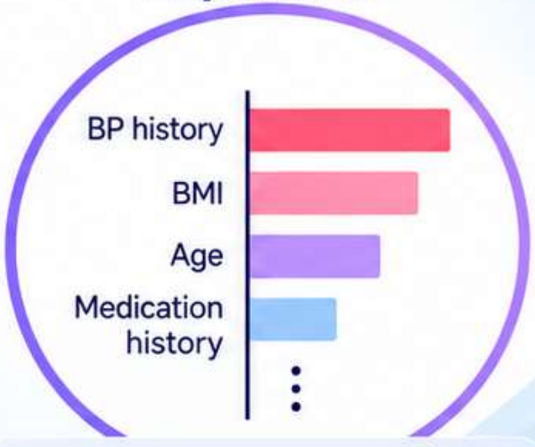
Model Risk Score



Model Score

Not a calibrated clinical probability.

SHAP Explanation



Understand key factors contributing to a higher risk.

VitalTwin — AI-Powered Clinical Digital Twin

The Danger of Isolated Clinical Snapshots

Why EHRs fall short

Isolated records can miss longitudinal patterns and provide an incomplete picture of a patient's health.

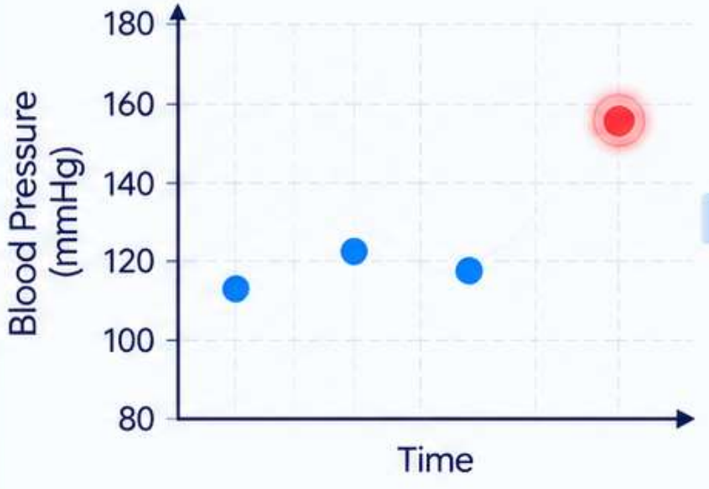
Why time matters

Repeated blood-pressure readings help reveal trends, variability, and changes over time.

Why explainability matters

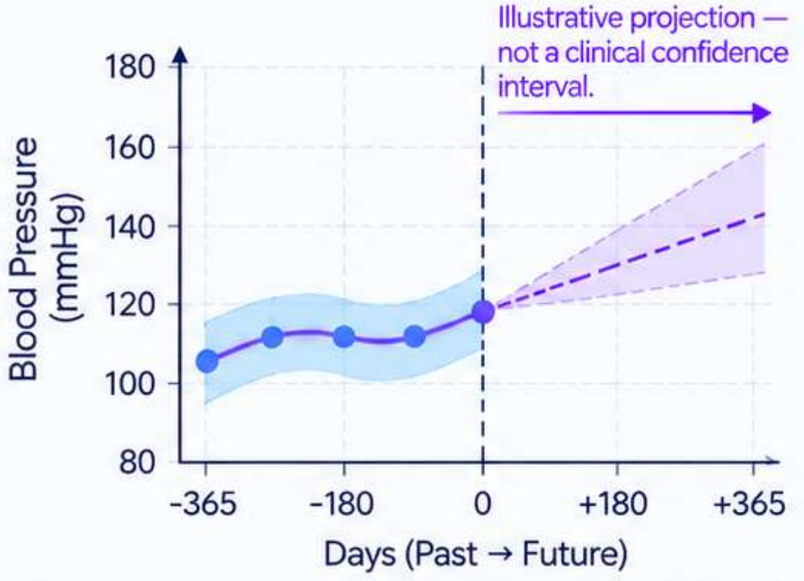
Model outputs need understandable explanations to be clinically actionable.

Traditional Snapshot-Based View



Individual readings provide **limited temporal context.**

Longitudinal Digital Twin View



A longitudinal patient state **summarizes patterns across repeated records.**

The VitalTwin Workflow

Core mechanism: Use a 365-day historical lookback to construct a longitudinal patient state and estimate incident hypertension risk over the next 365 days.

1 Patient Selection

Select a patient from the EHR dataset.

2 Historical Vitals & Clinical History



Collect 365 days of vitals, labs, medications, and clinical records.

3 Patient-State Construction



132 engineered features

Create a structured longitudinal patient state from historical data.

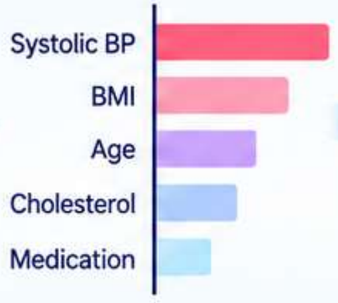
4 Model Score



Model score
Not a calibrated clinical probability.

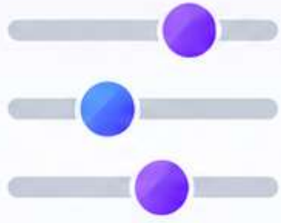
Estimate hypertension risk over the next 365 days using the existing XGBoost pipeline.

5 SHAP Explanation



Compute feature contributions using SHAP to explain the model score.

6 What-If Simulation



Explore model-based scenarios by modifying key inputs (e.g., blood pressure, BMI, medication adherence) and compare the resulting model score.

Engineered for Clinical Rigor and Scalability

5 Frontend: Visualization

Next.js Dashboard & Three.js

- Interactive dashboard
- 3D data visualization (Three.js)
- Patient timeline & insights
- What-if simulation UI





Frontend handles visualization only; all model inference is secured within the FastAPI backend.

4 Backend API

FastAPI (Python)

- Secure prediction endpoints
- Explanation (SHAP) endpoint
- What-if simulation endpoint
- Patient data & timeline endpoints
- Audit logging and access control

FastAPI
REST API

- **/predict** Model score
- **/explain** SHAP values
- **/simulate** What-if analysis
- **/patient** Patient timeline
- **/audit** Logging & security

3 Machine Learning Pipeline

Preprocessing + XGBoost

- Existing trained model
- Preprocessing with saved artifacts
- ColumnTransformer and Platt Scaling
- Model score and SHAP explanation

Preprocessing Pipeline
Saved artifacts (Transformations)

→

XGBoost Model
Existing trained model (no retraining in frontend)

→

Model Score
Hypertension risk score (model output) (not a calibrated clinical probability)

→

SHAP Explanation
Feature contributions for interpretability

2 Feature Engineering

Patient-State Construction

- 365-day lookback window
- Aggregate vitals, labs, medications, encounters
- 132 engineered features
- Prevent target leakage with strict temporal windows

365-Day Lookback Window
Collect patient records from past 365 days

→

Patient-State Construction
Create a structured patient state

→

132 Engineered Features
Temporal, statistical and clinical features

→

Feature Quality Checks
Handle missing data and apply constraints

1 Data

Synthea EHR Data

- Structured clinical records
- Vitals, labs, medications, encounters
- Temporal records for longitudinal analysis

Synthea EHR Dataset
Structured patient records (diagnoses, vitals, labs, medications, encounters, etc.)

→

365-Day Lookback
Use historical records from the past 365 days

→

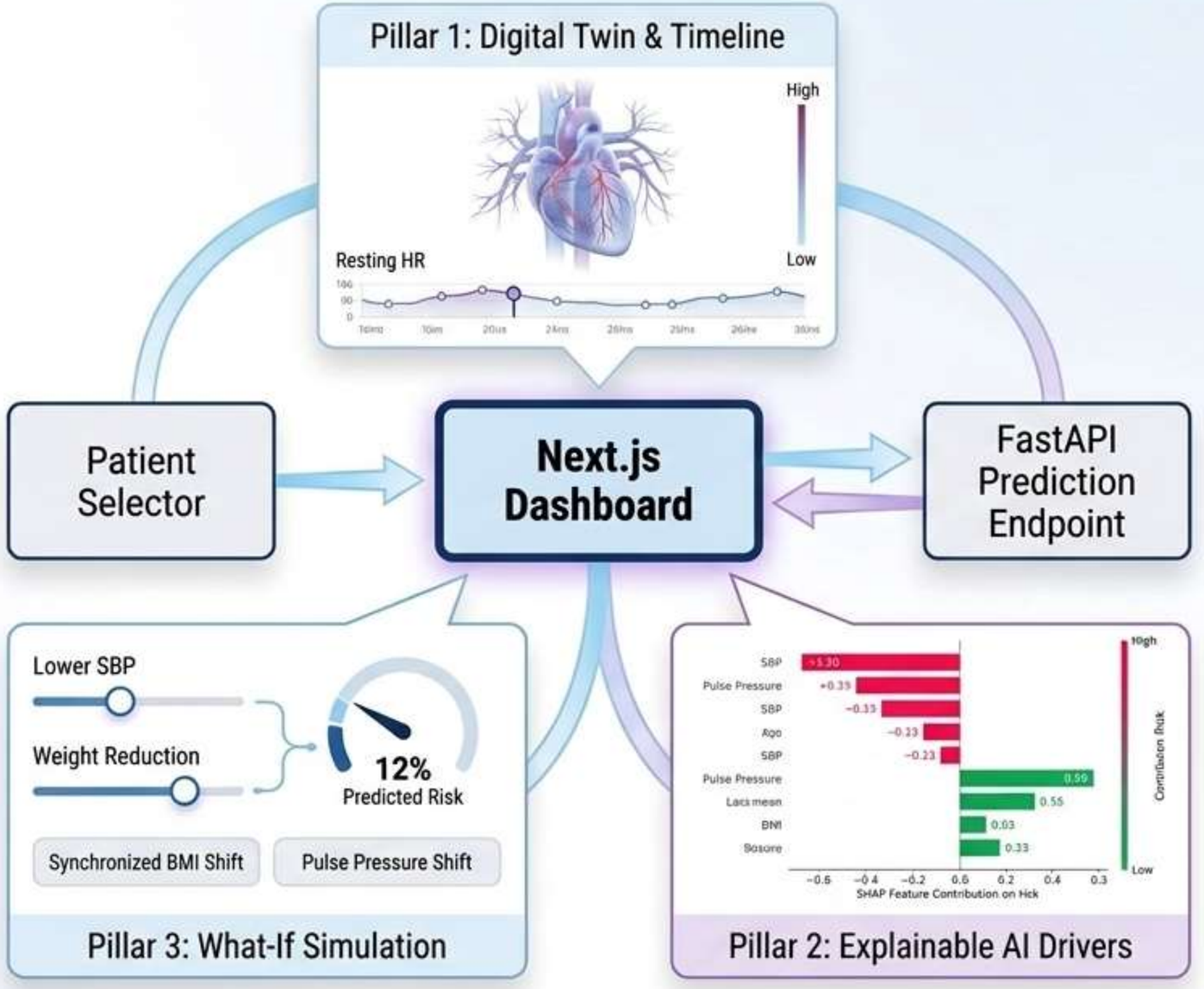
Wearable & Additional Indicators
Simulated wearable telemetry and derived indicators

Translating Complexity into Clinical Intuition

Pillar 1: Patient Digital Twin.
3D spatial physiological core synchronized with patient resting HR; longitudinal historical timeline.

Pillar 2: Explainable AI.
TreeExplainer SHAP feature contributions highlighting exact physiological risk drivers.

Pillar 3: What-If Simulation. Tests correlated clinical interventions (e.g., lower SBP triggers synchronized BMI and pulse pressure shifts).



Gemini Notebook

Honest Evaluation in Rare-Event Prediction

Implications of rarity
With a 1.46% base event rate, ROC-AUC can be misleadingly optimistic. PR-AUC reveals the true difficulty of rare-event prediction.

Threshold reality
A standard validation cutoff (0.7562) caught zero true positives, showing that alert-budgeting is required over arbitrary static thresholds.

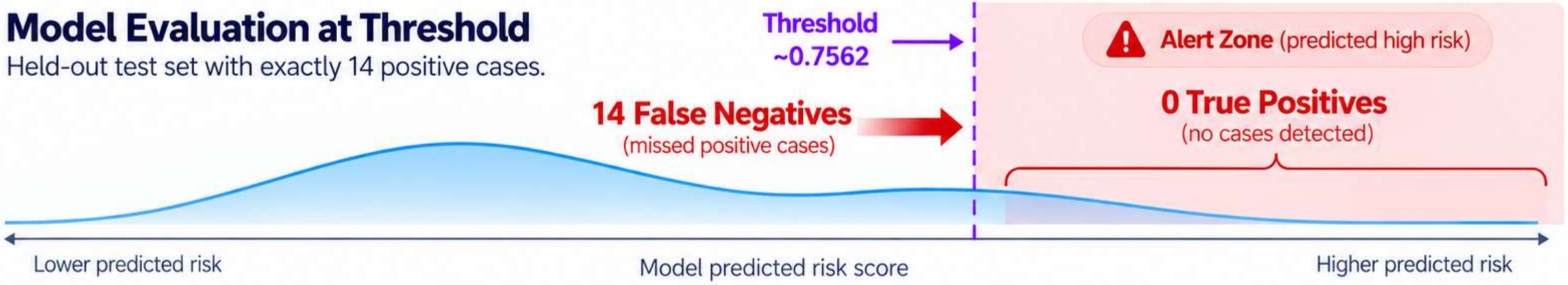
6,451
Supervised Records
132 Model Features

94
Positive Cases
~1.46% Positive Event Rate

ROC-AUC
0.8069
(95% CI: 0.691 – 0.908)

PR-AUC
0.0644
Compare against **1.46%** baseline due to rarity.

Model Evaluation at Threshold
Held-out test set with exactly 14 positive cases.



Synthetic-data evaluation; not clinically validated.

VitalTwin — AI-Powered Clinical Digital Twin

Implementation Roadmap and Future Development

Completed

1. Dataset Exploration & Audit
2. Longitudinal Patient-State Generation
3. Feature Engineering
4. Model Training & Evaluation
5. Phase 15: Inference Safeguards
6. Phase 16: Website & API Integration

Future

- External Clinical Validation
- Model Calibration & Threshold Evaluation
- Indian Population Adaptation
- Clinician-led Usability Testing

Frontend Next.js Container

User interface and visualization dashboard.

FastAPI Backend Server

Model inference, data processing, and API services.

Model Artifacts (Joblib)

Trained model and related assets.

Data (FHIR/JSON)

Processed clinical data in FHIR/JSON format.

Data (FHIR)

Source EHR data in FHIR format.

Lightweight, scalable infrastructure designed for low-latency clinical environments.

Scalable | Secure | Cloud-Ready

A Continuous Trajectory Toward Preventive Care

Longitudinal patient states, explainable model outputs, and interactive what-if simulations can support research into earlier hypertension risk identification.



Research prototype • Clinical validation and Indian-population evaluation are future steps.